

# A Wronskian Identity for Antisymmetrized Sine Sums: the surface-positivity problem as sine-series ratio monotonicity, and a counterexample to the naive monotonicity principle

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## Abstract

For real arrays  $(a_m)_{m \geq 1}$ ,  $(b_m)_{m \geq 1}$  (finitely supported, or summable with  $\sum m(|a_m| + |b_m|) < \infty$ ) we record the exact identity

$$\sum_{1 \leq m < n} (a_m b_n - a_n b_m) [(m-n) \sin((m+n)t) + (m+n) \sin((n-m)t)] = 2 [F'_a(t) F_b(t) - F_a(t) F'_b(t)],$$

where  $F_x(t) = \sum_m x_m \sin(mt)$ : the antisymmetrized double sum is a Wronskian of two sine series. The proof is two lines. As a consequence, the open surface-positivity problem of the companion  $\varphi$ -lemma note — the sign of exactly such a double sum built from modified-Bessel coefficient arrays — is *equivalent* to the strict monotonicity of a ratio of two explicit sine series, and high-precision numerics indicate that the true statement is global: the Wronskian is negative on all of  $(0, \pi)$ , not merely near  $t = \pi$ . One might hope this follows from the companion result that the coefficient ratio  $a_m/b_m$  is strictly increasing, in analogy with the Biernacki–Krzyż principle for power series. It does not: we give a hand-checkable counterexample —  $a = (3, 4, 3)$ ,  $b = (3, 2, 1)$ , coefficient ratio  $(1, 2, 3)$  strictly increasing,  $b$  strictly decreasing so that  $F_b > 0$  on  $(0, \pi)$  by the classical Fejér–Gronwall positivity theorem, yet the Wronskian at  $t = 2\pi/3$  equals  $3\sqrt{3}/2 > 0$  — so no analogue of Biernacki–Krzyż holds for sine series, and the surface problem cannot be closed by coefficient-ratio monotonicity alone. We state the resulting conjecture precisely, record the numerical phenomenology (near-perfect parity-mirror cancellation of relative size  $e^{-c\beta}$ ,  $c \approx 2.1$ , with a working-precision warning), and identify the structural route we consider most promising. The identity and the counterexamples are verified symbolically and in exact arithmetic; nothing beyond the identity and the counterexamples is claimed.

## 1 The identity

**Theorem 1** (Wronskian identity). *Let  $(a_m)$ ,  $(b_m)$  be real arrays, finitely supported (or with  $\sum_m m(|a_m| + |b_m|) < \infty$ , in which case all series below converge absolutely and uniformly). Set  $F_a(t) = \sum_{m \geq 1} a_m \sin(mt)$  and likewise  $F_b$ . Then, for all  $t \in \mathbb{R}$ ,*

$$\sum_{1 \leq m < n} (a_m b_n - a_n b_m) [(m-n) \sin((m+n)t) + (m+n) \sin((n-m)t)] = 2 [F'_a(t) F_b(t) - F_a(t) F'_b(t)]. \quad (1)$$

*Proof.* The product-to-sum formulas give, for any  $m, n$ ,

$$(m-n) \sin((m+n)t) + (m+n) \sin((n-m)t) = 2 [m \cos(mt) \sin(nt) - n \sin(mt) \cos(nt)] =: 2G(m, n),$$

and  $G$  is antisymmetric with  $G(n, n) = 0$ . Since  $(a_m b_n - a_n b_m)$  is antisymmetric as well,

$$\sum_{m < n} (a_m b_n - a_n b_m) 2G(m, n) = 2 \sum_{m, n} a_m b_n G(m, n) = 2 \left[ \sum_m a_m m \cos(mt) \sum_n b_n \sin(nt) - \sum_m a_m \sin(mt) \sum_n b_n n \cos(nt) \right]$$

which is  $2[F'_a F_b - F_a F'_b]$ .  $\square$

*Remark 2.* Equivalently: wherever  $F_b \neq 0$ , the left side of (1) equals  $2F_b^2 \cdot (F_a/F_b)'$ . Sign questions about the antisymmetrized double sum are exactly monotonicity questions about the ratio of the two sine series — with the Wronskian form remaining meaningful (and division-free) across zeros of  $F_b$ .

## 2 The surface-positivity problem, reduced

We recall the objects of the companion note [4] (which proved the coefficient-ratio monotonicity used below; the wider programme context is in [6]). For  $\beta > 0$  let  $r_k = I_{k+1}(\beta)/I_1(\beta)$  with  $I_\mu$  the modified Bessel function of the first kind, and for  $k \geq 1$

$$A_k = r_{k-1}^2 \sum_{j \in \{k-2, k\}, j \geq 0} (j+1) r_j^2, \quad B_k = k r_{k-1}^4, \quad c_{mn} = A_m B_n - A_n B_m.$$

The open positivity problem is  $W(t) < 0$  for  $t$  near  $\pi$ , where  $W$  is the left side of (1) with  $(a, b) = (A, B)$ . By Theorem 1,

$$W(t) = 2 [F'_A(t) F_B(t) - F_A(t) F'_B(t)], \quad F_A(t) = \sum_{m \geq 1} A_m \sin(mt), \quad F_B(t) = \sum_{m \geq 1} B_m \sin(mt).$$

Two facts sharpen the picture. First, the companion note proved  $c_{mn} < 0$  for all  $m < n$  (the coefficient ratio  $\varphi_m = A_m/B_m$  is strictly increasing) [4]. Second, high-precision numerics (mpmath, 30–100 significant digits, truncation-stable) indicate that the Wronskian is strictly negative on *all* of  $(0, \pi)$ , at every  $\beta$  tested ( $\beta \in \{0.5, 2, 8, 32\}$ , grids of  $t/\pi$  from 0.05 to 0.99), and that  $F_B > 0$  on  $(0, \pi)$ . The  $\varepsilon$ -window form of the original problem is thus the boundary shadow of a global statement, which we now record as the target.

*Conjecture 3* (global ratio monotonicity). For every  $\beta > 0$ :  $F_B > 0$  on  $(0, \pi)$ , and  $F_A/F_B$  is strictly decreasing on  $(0, \pi)$ ; equivalently  $W(t) < 0$  on  $(0, \pi)$ .

## 3 The naive principle fails

For power series with positive coefficients, the Biernacki–Krzyż principle [1] states that if  $a_m/b_m$  is increasing then  $\sum a_m x^m / \sum b_m x^m$  is increasing on  $(0, 1)$ . Given  $\varphi_m = A_m/B_m$  increasing and  $W < 0$  numerically, it is tempting to conjecture the sine-series analogue: *coefficient ratio increasing  $\Rightarrow$  ratio of sine series decreasing on  $(0, \pi)$*  (the sign flip suggested by the boundary behaviour at  $t = \pi$ ). This is false, even under strong positivity side conditions.

**Proposition 4** (counterexample, hand-checkable). *Let  $a = (3, 4, 3)$  and  $b = (3, 2, 1)$  (all further entries zero). Then: the coefficient ratio  $a_m/b_m = (1, 2, 3)$  is strictly increasing;  $b$  is strictly decreasing, so  $F_b(t) = 3 \sin t + 2 \sin 2t + \sin 3t > 0$  on  $(0, \pi)$  by the Fejér–Gronwall positivity theorem for sine polynomials with decreasing coefficients [2]; and yet at  $t = 2\pi/3$ ,*

$$F_a = -\frac{\sqrt{3}}{2}, \quad F_b = \frac{\sqrt{3}}{2}, \quad F'_a = \frac{7}{2}, \quad F'_b = -\frac{1}{2}, \quad F'_a F_b - F_a F'_b = \frac{7\sqrt{3}}{4} - \frac{\sqrt{3}}{4} = \frac{3\sqrt{3}}{2} > 0.$$

Hence  $F_a/F_b$  is strictly increasing at  $t = 2\pi/3$ : no Biernacki–Krzyż analogue holds for sine series.

*Proof.* At  $t = 2\pi/3$ :  $\sin t = \frac{\sqrt{3}}{2}$ ,  $\sin 2t = -\frac{\sqrt{3}}{2}$ ,  $\sin 3t = 0$ ;  $\cos t = -\frac{1}{2}$ ,  $\cos 2t = -\frac{1}{2}$ ,  $\cos 3t = 1$ . Then  $F_a = (3 - 4)\frac{\sqrt{3}}{2} = -\frac{\sqrt{3}}{2}$ ,  $F_b = (3 - 2)\frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$ ,  $F'_a = 3(-\frac{1}{2}) + 8(-\frac{1}{2}) + 9 = \frac{7}{2}$ ,  $F'_b = 3(-\frac{1}{2}) + 4(-\frac{1}{2}) + 3 = -\frac{1}{2}$ ; combine.  $\square$

An even smaller example, with the Wronskian form:  $a = (1, 2, 3, 4)$ ,  $b = (1, 1, 1, 1)$  gives  $F'_a F_b - F_a F'_b = 4$  at  $t = \pi/2$  (there  $F_b$  vanishes, so this violates  $W \leq 0$  though not ratio-monotonicity on a domain of positivity — which is why Proposition 4 carries the decreasing- $b$  condition).

*Remark 5* (the moral, in the spirit of the parity barrier). As with the parity barrier for decoupling inequalities [5], a bounded amount of structural data — here, the monotonicity of the coefficient ratio, even together with positivity of  $F_b$  — cannot certify the global sign. Whatever proves Conjecture 3 must consume more of the Bessel structure of  $(A_m, B_m)$  than the order of  $\varphi_m$ . This is consistent with the quantitative phenomenology: the even- and odd-parity halves of  $W(\pi - \varepsilon)$  cancel to relative accuracy  $\approx e^{-c\beta}$  with  $c \approx 2.03$ – $2.17$  (true totals e.g.  $-2.9 \cdot 10^{-8}$  at  $\beta = 8$ ;  $-5.0 \cdot 10^{-29}$  at  $\beta = 32$ ;  $-5.1 \cdot 10^{-95}$  at  $\beta = 100$ ), so any estimate that discards the exact structure loses the sign entirely; numerical exploration below  $\approx 2.2\beta + 20$  working digits sees pure rounding noise. The route we consider most promising:  $r_k^2$  and  $r_k^4$  are ratios of Bessel products with integral (Schläfli/heat-kernel) representations, so  $F_A$  and  $F_B$  admit kernel representations on the circle; a positive integral representation of  $-(F_A/F_B)'$  — or a total-positivity property of the array  $(A; B)$  against the sine kernel — would close Conjecture 3 where coefficient-order arguments provably cannot.

## 4 Verification

Theorem 1 was verified symbolically (sympy) for generic coefficients at  $K \leq 5$  (residual 0 after trigonometric normalisation) and at  $K = 7$  by exact rational substitution (residual  $< 10^{-35}$  at 40 digits, four random points), and numerically against the direct double sum for the Bessel arrays at  $K = 90$ ,  $\beta = 8$ , at  $t \in \{0.5, 1.5, 2.6, \pi - 0.1\}$  (ratio 1.0 to all displayed digits). The counterexample values of Proposition 4 are exact algebraic evaluations. The global sign scan of  $W$  and the parity-mirror measurements are as in the companion note. Scripts and sources are archived at

<https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME> (docs/BF2-ATTACK-NOTES.md, papers/wronskian-reduction).

A Lean 4 formalization of Theorem 1 (a finite antisymmetrization over `Finset` with the sine/cosine addition formulas) is the natural next target of the programme’s formalization satellite; it is *not* claimed here.

## Acknowledgements

This note is part of an audit-first research programme whose records, Lean sources and numerical suites are public at <https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME>. Nothing is claimed about Conjecture 3 beyond its equivalence to the original surface-positivity problem and the failure of the naive route; nothing is claimed about four-dimensional gauge theory.

## References

- [1] M. Biernacki and J. Krzyż, *On the monotonicity of certain functionals in the theory of analytic functions*, Ann. Univ. Mariae Curie-Skłodowska Sect. A **9** (1955), 135–147.
- [2] R. Askey and J. Steinig, *Some positive trigonometric sums*, Trans. Amer. Math. Soc. **187** (1974), 295–307.
- [3] F. W. J. Olver et al., *NIST Digital Library of Mathematical Functions*, §10.29, <https://dlmf.nist.gov/10.29>.
- [4] L. Eriksson, *A weighted Turán-type monotonicity lemma for modified Bessel functions via the calibrated Amos bound, with an application to the ordering of a surface expansion in two-dimensional lattice gauge theory*, July 2026; <https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME/tree/main/papers/phi-lemma>.
- [5] L. Eriksson, *Parity barriers for decoupling inequalities*, July 2026; <https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME/tree/main/papers/parity-barriers>.
- [6] L. Eriksson, *THE MASTER MAP: an audit-first navigation guide to the conditional construction of 4D  $SU(N)$  Yang–Mills with mass gap*, ai.viXra:2602.0096, and the programme repository <https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME>.